

PHPE 308M/PHIL 209F - Fairness

Instructor:	Eric Pacuit (pacuit.org)
Semester:	Fall 2025
Email:	epacuit@umd.edu
Course Website:	https://umd.instructure.com/courses/1391244
Class Times:	MW 2:00pm - 3:15pm
Location:	JMZ 0125
Office Hours:	Tuesdays, 11am - 12pm
Office:	Skinner 1103A

Course Description

Fairness is a recurring theme in game theory, social choice theory, political philosophy, and ethics. Today, as AI systems influence critical decisions—from loan approvals to criminal risk assessments—it is crucial to understand when algorithms are fair. This course is divided into four parts, each focused on different ways that fairness considerations influence individual and group decision-making:

1. Fairness norms and social preferences;
2. Fair division;
3. Fair representation; and
4. Algorithmic fairness.

This course equips students to identify fairness trade-offs, analyze competing fairness criteria, and evaluate both human and algorithmic decision-making systems.

Course Structure

This course will have weekly lectures on Mondays and Wednesdays 2:00 pm - 3:15 pm. During the lectures, we will introduce the material for the week, discuss the readings, and work together on participation questions.

Required Resources

- Course Website: <https://umd.instructure.com/courses/1391244>
- There is no textbook to purchase for this course. The course will be based on readings from various textbooks and journal articles. The relevant material will be made available on the course website.
- We will PollEverywhere this semester. This will be used for short quizzes and surveys given during the lectures. You can sign-up for PollEverywhere for free by following this link https://PollEv.com/epacuit/register?group_key=CzVqi3EcbxMoQXErccMFgG8qm.

Grading Policy

The course requirements are:

- **Participation:** There will be a number of short quizzes given periodically during the semester. Typically, these short quizzes will consist of 1 question that is given during class using PollEverywhere, but some lectures may have more than 1 question. **Make-up quizzes will not be offered.** I will drop the lowest 5-10% of the quizzes (so you can miss some of the questions without losing any points).
- **Weekly reflection:** Students must submit a weekly reflection on ELMS. Each week, you will be asked to describing one thing that surprised/confused/interested you from the readings and/or the lectures during week, or an example you observed of an interaction that illustrates some issue discussed that week. Your reaction should be approximately 200 words. Each weekly reflection is worth 5 points and is **due Friday at 11:59pm**. You will be graded, in part, on how it relates to the material discussed that week.
- **Problem Sets:** There will be 4 problem sets (after each part). Problem sets will be submitted through the course website. You can use your notes, the readings, and the online textbook, but you should not discuss your answers with your classmates or use any AI tools, such as ChatGPT, to answer these questions.
- **Final exam:** There will be an in-person final exam given during finals week. Consult <https://registrar.umd.edu/registration/register-classes/final-exams/fall> for more information about the time and location of the final exam.

Grades will be assigned according to the following weights:

Participation	25%
Weekly Reflection	30%
Problem Sets	30%
Final Exam	15%

Your final grade may be curved, but the final grade cutoffs are typically as follows:

A+	> 100%	B+	88%	C+	78%	D+	68%	F < 60%
A	92%	B	82%	C	72%	D	62%	
A-	90%	B-	80%	C-	70%	D-	60%	

Tentative Syllabus

Below is a tentative syllabus for this semester. This is subject to change. Please consult the course website for an up-to-date overview of the material discussed in the course.

Part 1: Fairness norms and social preferences

Ultimatum Game (Wednesday 9/3, Monday 9/8, Wednesday, 9/10)

Colin Camerer and Richard H. Thaler (1995). Anomalies: Ultimatums, Dictators and Manners, *Journal of Economic Perspectives*, 9(2), pp. 209 - 219.

Cristina Bicchieri and Jiji Zhang (2012). An Embarrassment of Riches: Modeling Social Preferences in Ultimatum Games, in *Philosophy of Economics: A Handbook of the Philosophy of Science* pp. 577 - 595. Sections 1 & 2

Armin Falk, Ernst Fehr and Urs Fischbacher (2003). On the Nature of Fair Behavior, *Economic Inquiry*, 41, pp. 20-26.

Nash Bargaining (Monday 9/15, Wednesday 9/17, Monday 9/22, Wednesday 9/24)

Brian Skyrms (2014). Chapter 1: Sex and Justice (pp. 1-22) and Chapter 2: Fairness and Commitment (pp. 23-42), in *Evolution of the Social Contract: Second Edition*, Cambridge University Press.

J. McKenzie Alexander and B. Skyrms (1999). Bargaining with Neighbors: Is Justice Contagious? *Journal of Philosophy*, 96(11), pp. 588-598.

Unfairness (Wednesday 9/24, Monday 9/29, Wednesday 10/1)

Cailin O'Connor (2024). Why Natural Social Contracts are Not Fair, in *New Approaches to Social Contract Theory: Liberty, Equality, Diversity, and the Open Society*, Oxford University Press.

Cailin O'Connor (2019). Chapter 5: Power and the Evolution of Unfairness (pp. 105 - 132, in *Origins of Unfairness*, Oxford University Press.

Additional reading

Martin A. Nowak, Karen M. Page, Karl Sigmund (2000). Fairness Versus Reason in the Ultimatum Game, *Science*, 289, pp. 1773-1775.

Kevin Zollman (2008). Explaining fairness in complex environments, *Philosophy, Politics, and Economics*, 7(1), pp. 81 - 98.

Part 2: Fair division

Fair division of indivisible goods (Monday 9/29, Wednesday 10/1)

H. Peyton Young (1994). Chapter 1: Overview (pp. 3 - 19) and Chapter 2: Equity and Priority (pp. 20 - 41) in *Equity in Theory and Practice*, Princeton University Press

Steven Brams, Paul Edelman, Peter Fishburn (2001). Paradoxes of Fair Division, *The Journal of Philosophy*, 98(6), pp. 300-314.

Steven J. Brams, D. Marc Kilgour, Christian Klamler, and Fan Wei (2023). Two-Person Fair Division of Indivisible Items - Bentham vs. Rawls on Envy, *Journal of Philosophy*, 120 (8), pp. 441-456 (2023)

Fair division of divisible goods (Monday 10/6, Wednesday 10/8)

Steven J. Brams and Alan D. Taylor (1996). Chapter 1: Proportionality for $n = 2$ (pp. 6-29) and Chapter 4: Envy-freeness and equitability for $n = 2$ (pp. 65 - 94) in *Fair Division: From Cake-Cutting to Dispute Resolution*, Cambridge University Press.

Steven Brams, Michael A. Jones and Christian Klamler (2006). Better Ways to Cut a Cake. *Notices of the AMS*, 53:11, pp. 1314-1321.

Lotteries (Wednesday 10/15, Monday 10/20, Wednesday 10/22)

John Broome (1990). Fairness, *Proceedings of the Aristotelian Society*, 91, 87-101.

A.C. Paseau and Ben Saunders (2015). Fairness and Aggregation. *Utilitas* 27 (4), pp. 460-469.

Kfir Eliaz and Ariel Rubinstein (2014). On the fairness of random procedures, *Economics Letters*, 123, pp. 168 - 170

Additional reading

Hal R. Varian (1975). Distributive Justice, Welfare Economics, and the Theory of Fairness, *Philosophy & Public Affairs*, 4(3), pp. 223 - 247.

Benjamin L. Curtis (2014). To be fair, *Analysis*, 74(1), pp. 47-57.

Ariel D. Procaccia (2016). Cake Cutting Algorithms, in *Handbook of Computational Social Choice* (Brandt, Conitzer, Endriss, Lang, and Procaccia, eds.), chapter 13.

Part 3: Fair representation

Gerrymandering (Monday 10/27, Wednesday 10/29)

Moon Duchin (2022). Chapter 1: Introduction in *Political Geometry: Rethinking Redistricting in the US with Math, Law, and Everything In Between*, Moon Duchin, Olivia Walch (eds.), Birkhuaser, pp. 29-35.

Steven Brams (2020). Making Partisan Gerrymandering Fair: One Old and Two New Methods. *Social Science Quarterly*, 101(1), pp. 68 - 72.

Mira Bernstein and Olivia Walch (2022). Chapter 2: Measuring partisan fairness in *Political Geometry: Rethinking Redistricting in the US with Math, Law, and Everything In Between*, Moon Duchin, Olivia Walch (eds.), Birkhuaser.

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Proportional Representation and Multiwinner Voting (Monday 11/3, Wednesday 11/5)

Gerdus Benade, Ruth Buck, Moon Duchin, Moon, Dara Gold, and Thomas Weighill (2021). Ranked Choice Voting and Proportional Representation, Available on SSRN: <http://dx.doi.org/10.2139/ssrn.3778021>.

P. Faliszewski, P. Skowron, A. Slinko, and N. Talmon (2017). Multiwinner Voting: A New Challenge for Social Choice Theory. In U. Endriss (ed.), *Trends in Computational Social Choice*. AI Access.

Additional reading

Z. Landau, O. Reid and I. Yershov (2009). A fair division solution to the problem of redistricting. *Social Choice and Welfare*, 32(3), pp. 479 - 492.

Bailey Flanigan, Paul Gözl, Anupam Gupta, Brett Hennig, and Ariel D. Procaccia (2021). Fair algorithms for selecting citizens' assemblies, *Nature*, 596, pp. 548-552.

Ariel Procaccia (2022). Citizens' Assemblies Are Upgrading Democracy: Fair Algorithms Are Part of the Program Math, *Scientific American*.

Gerdus Benadé, Ariel D. Procaccia, and Jamie Tucker-Foltz (2023). You Can Have Your Cake and Redistrict It Too, in *Proc. 24th ACM Conference on Economics and Computation*.

Michel L. Balinski and H. Peyton Young (1975). The Quota Method of Apportionment, *The American Mathematical Monthly*, 82(7), pp. 701-730.

Geoffrey Grimmett (2004). Stochastic Apportionment, *The American Mathematical Monthly*, 111(4), pp. 299 - 307.

Part 4: Algorithmic fairness

AI and Fairness (Monday 11/10, Wednesday 11/12)

Jana Schaich Borg, Walter Sinnott-Armstrong, Vincent Conitzer (2024). *Moral AI And How We Get There*, Ch 4: Can AI be fair?, Penguin Books

John W. Patty and Elizabeth Maggie Penn (2022). Algorithmic Fairness and Statistical Discrimination, *Philosophy Compass*, 18(1), e12891.

Fairness Criteria (Monday 11/17, Wednesday 11/19, Monday 11/24, Monday 12/1, Wednesday 12/3)

Deborah Hellman (2025). Algorithmic Fairness, The Stanford Encyclopedia of Philosophy, <https://plato.stanford.edu/archives/fall2025/entries/algorithmic-fairness/>.

Brian Hedden (2021). On statistical criteria of algorithmic fairness, *Philosophy and Public Affairs*, 49(2), pp. 209-231

Benjamin Eva (2022). Algorithmic Fairness and Base Rate Tracking, *Philosophy and Public Affairs*, 50(2), pp. 239-266.

Rush T. Stewart (2022). Identity and the Limits of Fair Assessment, *Journal of Theoretical Politics*, 34(3), pp. 415-442.

Seth Lazar and Jake Stone (2024). On the site of predictive justice, *Noûs*, 58(3), pp. 563 - 847.

Solon Barocas, Moritz Hardt, Arvind Narayanan (2023). *Fairness and machine learning: Limitations and opportunities* (selected readings), <https://fairmlbook.org/>.

Human and AI decision-making (Monday 12/8, Wednesday 12/10)

Ruth Chang (2024). Human in the Loop!, in David Edmonds (ed.), *AI Morality*, Oxford Academic.

Jakob Schoeffer, Maria De-Arteaga, and Niklas Kühl (2024). Explanations, Fairness, and Appropriate Reliance in Human-AI Decision-Making, <https://arxiv.org/abs/2209.11812>

Additional Reading

Sina Fazelpour and David Danks (2021). Algorithmic bias: Senses, sources, solutions, *Philosophy Compass*, 16:e12760.

Will Fleisher (2021). What's Fair about Individual Fairness? In Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society (AIES '21).

Sam Corbett-Davies, Johann D. Gaebler, Hamed Nilforoshan, and Ravi Shroff (2023). The Measure and Mismeasure of Fairness, *Journal of Machine Learning Research*, 24, pp. 1-117, Sections 1 & 2.

Zeyu Tang, Jiji Zhang, and Kun Zhang (2023). What-is and How-to for Fairness in Machine Learning: A Survey, Reflection, and Perspective, *ACM Computing Surveys*, 55(13), pp. 1 - 37.

I. Glenn Cohen, Boris Babic, Sara Gerke, Qiong Xia, Theodoros Evgeniou, and Klaus Wertenbroch (2023). How AI can learn from the law: putting humans in the loop only on appeal. *npj Digital Medicine*, 6, 160.

Evdoxia Taka, Yuri Nakao, Ryosuke Sonoda, Takuya Yokota, Lin Luo, and Simone Stumpf (2018). Human-in-the-loop Fairness: Integrating Stakeholder Feedback to Incorporate Fairness, Perspectives in Responsible AI. *Journal of the ACM*, 37, 4, Article 111.

Course Policies & Resources

- All course related policies and resources for undergraduate students:
<http://www.ugst.umd.edu/courserelatedpolicies.html>.
- Campus land acknowledgement statement
<https://admissions.umd.edu/student/land-acknowledgement>

Communication about this Course I will use ELMS announcements to convey important information, and students are responsible for keeping their email address up to date, and must ensure that forwarding to another email address functions properly. Failure to check email, errors in forwarding, and returned email are the responsibility of the student, and do not constitute an excuse for missing announcements or deadlines.

Communication with Peers With a diversity of perspectives and experience, we may find ourselves in disagreement and/or debate with one another. As such, it is important that we agree to conduct ourselves in a professional manner and that we work together to foster and preserve a virtual classroom environment in which we can respectfully discuss and deliberate controversial questions.

I encourage you to confidently exercise your right to free speech—bearing in mind, of course, that you will be expected to craft and defend arguments that support your position. Keep in mind, that free speech has its limit and this course is NOT the space for hate speech, harassment, and derogatory language. I will make every reasonable attempt to create an atmosphere in which each student feels comfortable voicing their argument without fear of being personally attacked, mocked, demeaned, or devalued.

Any behavior (including harassment, sexual harassment, and racially and/or culturally derogatory language) that threatens this atmosphere will not be tolerated. Please alert me immediately if you feel threatened, dismissed, or silenced at any point during our semester together and/or if your engagement in discussion has been in some way hindered by the learning environment.

Names/Pronouns and Self-Identifications The University of Maryland recognizes the importance of a diverse student body, and we are committed to fostering inclusive and equitable classroom environments. I invite you, if you wish, to tell us how you want to be referred to both in terms of your name and your pronouns (he/him, she/her, they/them, etc.). The pronouns someone indicates are not necessarily indicative of their gender identity. Visit the campus' list of Gender Inclusive Resources to learn more.

Additionally, how you identify in terms of your gender, race, class, sexuality, religion, and dis/ability, among all aspects of your identity, is your choice whether to disclose (e.g., should it come up in classroom conversation about our experiences and perspectives) and should be self-identified, not presumed or imposed. I will do my best to address and refer to all students accordingly, and I ask you to do the same for all of your fellow Terps.

Academic Integrity The University's Code of Academic Integrity is designed to ensure that the principles of academic honesty and integrity are upheld. In accordance with this code, academic dishonesty will not be tolerated. Please ensure that you fully understand this code and its implications because all acts of academic dishonesty will be dealt with in accordance with the provisions

of this code. All students are expected to adhere to this Code. It is your responsibility to read it and know what it says, so you can start your professional life on the right path.

It is important to note that course assistance websites, such as CourseHero, or AI-generated content are not permitted sources, unless the instructor explicitly gives permission. Material taken or copied from these sites can be deemed unauthorized material and a violation of academic integrity. These sites offer information that may be inaccurate or biased and that shortcut the learning process, particularly the critical thinking steps necessary for college-level assignments.

Additionally, it is understandable that students may use a variety of online or virtual forums for course-wide discussion (e.g., Group lists or chats). Collaboration in this way regarding concepts discussed in this course is permissible. However, collaboration on graded assignments is strictly prohibited unless otherwise stated. Examples of prohibited collaboration include: asking classmates for answers on quizzes or exams, asking for access codes to clicker polls, etc.

Finally, on each exam or assignment you must write out and sign the following pledge:

“I pledge on my honor that I have not given or received any unauthorized assistance on this exam/assignment.”

Please visit the Office of Undergraduate Studies’ full list of campus-wide policies and follow up with me if you have questions.

Accessibility and Disability Services The University of Maryland is committed to creating and maintaining a welcoming and inclusive educational, working, and living environment for people of all abilities. The University of Maryland is also committed to the principle that no qualified individual with a disability shall, on the basis of disability, be excluded from participation in or be denied the benefits of the services, programs, or activities of the University, or be subjected to discrimination. The Accessibility & Disability Service (ADS) provides reasonable accommodations to qualified individuals to provide equal access to services, programs and activities. ADS cannot assist retroactively, so it is generally best to request accommodations several weeks before the semester begins or as soon as a disability becomes known. Any student who needs accommodations should contact me as soon as possible so that I have sufficient time to make arrangements.

For assistance in obtaining an accommodation, contact Accessibility and Disability Service at 301-314-7682, or email them at adsfrontdesk@umd.edu.

Student Resources and Services Taking personal responsibility for your own learning means acknowledging when your performance does not match your goals and doing something about it. I hope you will come talk to me so that I can help you find the right approach to success in this course, and I encourage you to visit UMD’s Student Academic Support Services website to learn more about the wide range of campus resources available to you.

In particular, everyone can use some help sharpening their writing skills (and improving their grade) by visiting UMD’s Writing Center and schedule an appointment. For consulting and training in oral communication, please visit UMD’s Oral Communication Center.

You should also know there are a wide range of resources to support you with whatever you might need (UMD’s Student Resources and Services website may help). If you feel it would be helpful to have someone to talk to, visit UMD’s Counseling Center or one of the many other mental health resources on campus.

Notice of Mandatory Reporting Notice of mandatory reporting of sexual assault, sexual harassment, interpersonal violence, and stalking: As a faculty member, I am designated as a “Responsible University Employee,” and I must report all disclosures of sexual assault, sexual harassment, interpersonal violence, and stalking to UMD’s Title IX Coordinator per University Policy on Sexual Harassment and Other Sexual Misconduct.

If you wish to speak with someone confidentially, please contact one of UMD’s confidential resources, such as CARE to Stop Violence (located on the Ground Floor of the Health Center) at 301-741-3442 or the Counseling Center (located at the Shoemaker Building) at 301-314-7651.

You may also seek assistance or supportive measures from UMD’s Title IX Coordinator, Angela Nastase, by calling 301-405-1142, or emailing titleIXcoordinator@umd.edu.

To view further information on the above, please visit the Office of Civil Rights and Sexual Misconduct’s website.

Basic Needs Security If you have difficulty affording groceries or accessing sufficient food to eat every day, or lack a safe and stable place to live, please visit UMD’s Division of Student Affairs website for information about resources the campus offers you and let me know if I can help in any way.

Veteran Resources UMD provides some additional supports to our student veterans. You can access those resources at the office of Veteran Student Life. Veterans and active duty military personnel with special circumstances (e.g., upcoming deployments, drill requirements, disabilities) are welcome and encouraged to communicate these, in advance if possible, to the instructor.

Technology Policy Please refrain from using cellphones, laptops, and other electronic devices during class sessions unless we have designated such use as part of a class exercise. Further, video recording of class sessions or activities is prohibited without advance permission from the professor.

Participation

- Given the interactive style of this class, attendance will be crucial to note-taking and thus your performance in this class. Attendance is particularly important also because class discussion will be a critical component for your learning.
- Each student is expected to make substantive contributions to the learning experience, and attendance is expected for every session.
- Students with a legitimate reason to miss a live session should communicate in advance with the instructor, except in the case of an emergency.
- Students who miss a live session are responsible for learning what they miss from that session.
- Additionally, students must complete all readings and assignments in a timely manner in order to fully participate in class.

Course Evaluation Please submit a course evaluation through Student Feedback on Course Experiences in order to help faculty and administrators improve teaching and learning at Maryland. All information submitted to Course Experiences is confidential. Campus will notify you when Student Feedback on Course Experiences is open for you to complete your evaluations at the end of the semester. Please go directly to the Student Feedback on Course Experiences to complete your evaluations. By completing all of your evaluations each semester, you will have the privilege of accessing through Testudo the evaluation reports for the thousands of courses for which 70% or more students submitted their evaluations.

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